**Test of Mathematics for University Admission**

# TMUA October 2026

## Paper 1

Applications of Mathematical Knowledge

| | |
|---|---|
| **Time allowed** | 75 minutes |
| **Questions** | 20 |
| **Marks** | 20 |

### Instructions

- Choose one answer for each question.
- Each correct answer is worth one mark. There is no penalty for an incorrect answer.
- Calculators and formula booklets must not be used.
- Use the space in this booklet for rough work. An answer sheet for both papers is provided at the end.
- Allow 75 minutes for this paper, followed by 75 minutes for Paper 2.

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**About this compilation**

An unofficial reconstruction compiled from the two supplied documents. All 27 questions from the base document are retained, with 12 additional questions in Paper 1 and one in Paper 2. Some answer options have been constructed where the source did not provide them; clear transcription errors have been corrected. Question order is editorial and is not claimed to match the original examination.

## Question 1

For $-\frac{\pi}{2} \leq x < \frac{3\pi}{2}$, what is the sum of all solutions to the equation

$$\tan^3 x - \frac{2}{\sqrt{3}} \tan^2 x = \tan x \,?$$

**A** $\frac{\pi}{6}$

**B** $\frac{5\pi}{3}$

**C** $\frac{7\pi}{3}$

**D** $\frac{8\pi}{3}$

**E** $\frac{10\pi}{3}$

**F** $\frac{7\pi}{2}$

**G** $4\pi$

## Question 2

What is the complete range of values of $k$, where $k$ is a positive real number, for which the equation

$$2\sin(k\theta) = 1$$

has exactly five solutions in $0 \leq \theta \leq \pi$?

**A** $\frac{13}{6} \leq k < \frac{17}{6}$

**B** $2\frac{1}{6} \leq k < 2\frac{5}{6}$

**C** $4 \leq k < 5$

**D** $4\frac{1}{6} < k < 4\frac{5}{6}$

**E** $4\frac{1}{6} < k \leq 4\frac{5}{6}$

**F** $4\frac{1}{6} \leq k < 4\frac{5}{6}$

**G** $4\frac{1}{6} \leq k \leq 4\frac{5}{6}$

**H** $4\frac{1}{6} \leq k < 6\frac{1}{6}$

**I** $5\frac{1}{6} \leq k < 5\frac{5}{6}$

## Question 3

Points $P(1, 35)$ and $Q(4, 80)$ lie on the curve $y = 36x - x^3$. A point $R$ lies on the curve between $P$ and $Q$ and is such that the area of triangle $PQR$ is greatest.

What is the $x$-coordinate of $R$?

**A** $\frac{5}{2}$

**B** $\sqrt{7}$

**C** $3$

**D** $2\sqrt{3}$

**E** $\frac{7}{2}$

## Question 4

A function is defined by $f(x) = x^3 + 5x^2 + 3x + c$, where $c$ is a real constant. Given that $f'(x)$ shares a linear factor with $f(x)$, what is the sum of all possible values of $c$?

**A** $-\frac{230}{27}$

**B** $-9$

**C** $-\frac{1}{27}$

**D** $-\frac{10}{27}$

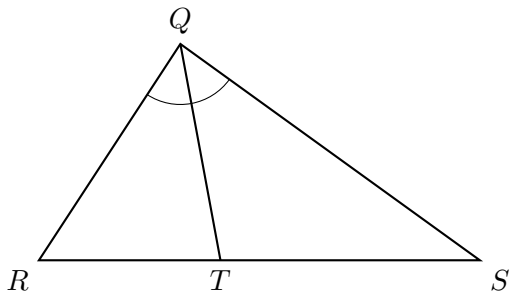
**E** $-\frac{1}{9}$

**F** $-10$

**G** $-27$

## Question 5

The diagram shows triangle $QRS$. The angles $RQT$ and $TQS$ are equal.




Which of the following is equal to $\frac{RT}{TS}$?

**A** $\frac{\sin S}{\sin R}$

**B** $\frac{\sin R}{\sin S}$

**C** $\frac{\tan S}{\tan Q}$

**D** $\frac{\tan Q}{\tan R}$

**E** $\frac{\cos S}{\cos R}$

**F** $\frac{\cos R}{\cos S}$

**G** $\frac{\tan S}{\tan R}$

## Question 6

A translation by $\begin{pmatrix} k \\ k \end{pmatrix}$ is applied to the curve with equation

$$y = x^3 - \frac{5}{4}x.$$

Which of the following gives the complete set of values of $k$ for which the translated curve has at least one point in common with the original curve?

**A** $-\sqrt{5} \leq k \leq \sqrt{5}$

**B** $-\sqrt{5} < k < \sqrt{5}$

**C** $-3 \leq k < 0$

**D** $-3 \leq k \leq 3$

**E** $k \geq 3$

**F** $k \geq \sqrt{5}$

**G** $3 \leq k \leq 9$

## Question 7

A polynomial has the form $x^3 + px^2 + qx - 27$. When it is divided by $(x + 2)$ it leaves a remainder of $-27$. When it is divided by $(x - 1)$ it leaves a remainder of $-15$.

What is the value of $2p + 13q$?

**A** 10

**B** 88

**C** 112

**D** 308

**E** 304

**F** 46

**G** 78

## Question 8

A circle $C$ has radius 4 and is tangent to the $x$-axis at $(2, 0)$. The line with equation $y = \sqrt{3}x - 2\sqrt{3}$ intersects $C$ at two distinct points $P$ and $Q$.

What is the length of the minor arc $PQ$?

**A** $\frac{2\pi}{3}$

**B** $\frac{4\pi}{3}$

**C** $4\sqrt{3}$

**D** $\frac{8\pi}{3}$

**E** $4\pi$

**F** $\frac{16\pi}{3}$

**G** $8\pi$

## Question 9

What is the value of

$$S = \sqrt{10} + \sqrt{100} + \sqrt{1000} + \cdots + \sqrt{1\,000\,000\,000}\,?$$

**A** $1111(\sqrt{10}+1)-1$

**B** $11111\sqrt{10}+1111$

**C** $11111(\sqrt{10}+1)$

**D** $11111(\sqrt{10}+1)-1$

**E** $11110(\sqrt{10}+1)$

## Question 10

A differentiable function satisfies

$$f'(x) = (x-1)(x-2)(x-3).$$

Each of the equations $f(x) = a$ and $f(x) = b$ has exactly three distinct real solutions.

How many distinct real solutions does

$$f(x) = \frac{a+b}{2}$$

have?

**A** 0

**B** 1

**C** 2

**D** 3

**E** 4

**F** 6

## Question 11

Real numbers $x$ and $y$ satisfy

$$\sqrt{y} + \sqrt{y+1} = \left(\sqrt{x} + \sqrt{x+1}\right)^2.$$

Which of the following gives $y$ explicitly as a function of $x$, together with the complete domain of $x$?

**A** $y = x(x+1), \quad x \geq 0$

**B** $y = 2x(x+1), \quad x \geq 0$

**C** $y = 4x(x+1), \quad x \geq 0$

**D** $y = 4x(x+1), \quad x \geq -1$

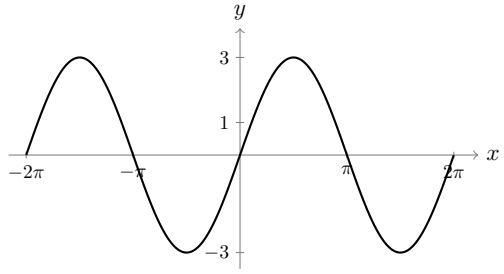
**E** $y = (2x+1)^2, \quad x \geq 0$

## Question 12

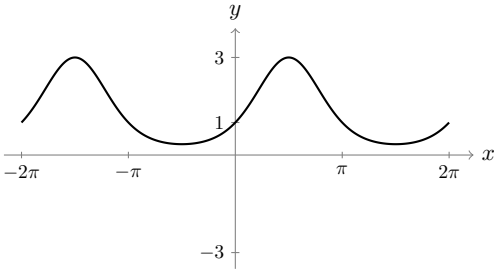
Which of the following graphs represents $y = 3^{\sin x}$?

Each graph is shown for $-2\pi \leq x \leq 2\pi$.

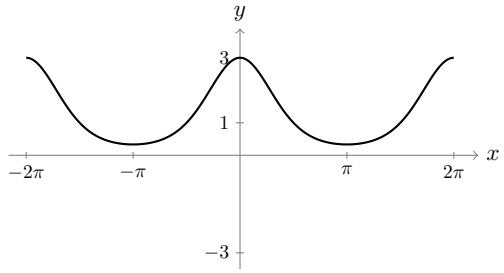
**A**



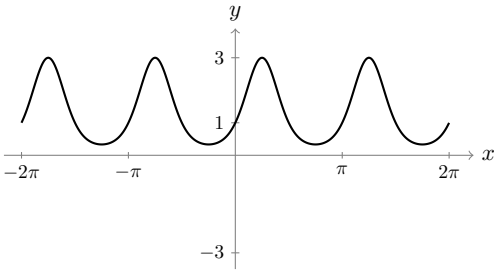

**B**



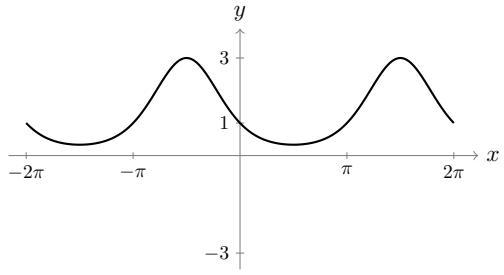

**C**




**D**




**E**



## Question 13

The graph of

$$y = 2^{|\log_2 x|}$$

is drawn for $x > 0$.

Which row gives the correct expression for $y$ on each of the indicated intervals?

| | $0 < x < 1$ | $x \geq 1$ |
|---|---|---|
| **A** | $x$ | $x$ |
| **B** | $x$ | $\frac{1}{x}$ |
| **C** | $\frac{1}{x}$ | $\frac{1}{x}$ |
| **D** | $\frac{1}{x}$ | $x$ |
| **E** | $-\frac{1}{x}$ | $x$ |

## Question 14

How many distinct real solutions does the equation

$$\cos^2(x^2 + x - 12) = 1$$

have in the interval $-\pi < x < \pi$?

**A** 3

**B** 4

**C** 5

**D** 6

**E** 7

**F** 8

## Question 15

Consider the following functions on $2 \leq x \leq 2.5$:

I $f(x) = |\cos x|$

II $f(x) = -\tan x$

III $f(x) = \log_{1/3} x$

For which functions does the trapezium rule give an underestimate of

$$\int_2^{2.5} f(x)\,dx\,?$$

**A** None of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

## Question 16

The graph of

$$y = \big|x - |x - 1|\big|$$

is drawn for all real $x$.

Which of the following gives its $x$-intercept, its $y$-intercept and its equation for $x \geq 1$?

| | $x$-intercept | $y$-intercept | Equation for $x \geq 1$ |
|---|---|---|---|
| **A** | $(1, 0)$ | $(0, 1)$ | $y = 1$ |
| **B** | $(\frac{1}{2}, 0)$ | $(0, -1)$ | $y = 1$ |
| **C** | $(\frac{1}{2}, 0)$ | $(0, 1)$ | $y = 2x - 1$ |
| **D** | $(-\frac{1}{2}, 0)$ | $(0, 1)$ | $y = 1$ |
| **E** | $(\frac{1}{2}, 0)$ | $(0, 1)$ | $y = 1$ |

## Question 17

For a real number $x$, let $\{x\}$ denote its fractional part, so that $\{x\} = x - \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to $x$.

What is the value of

$$\int_0^5 \big(2\{x\} - x\big)\, dx\,?$$

**A** $-\dfrac{25}{2}$

**B** $-10$

**C** $-\dfrac{15}{2}$

**D** $-5$

**E** $0$

**F** $\dfrac{15}{2}$

## Question 18

Let $f(x) = x^2 - 6x + 9$.

How many distinct real solutions does

$$f\big(f(f(x))\big) = 0$$

have?

**A** 1

**B** 2

**C** 3

**D** 4

**E** 6

**F** 8

## Question 19

How many distinct real roots does the equation

$$\left(\left(\left((x-1)^2-1\right)^2-1\right)^2-1\right)^2=1$$

have?

**A** 3

**B** 4

**C** 5

**D** 6

**E** 7

## Question 20

Let $a$, $b$ and $c$ be real numbers with $a < b < c$.

Consider the following statements:

I $b^2 < a^2$

II $b^2 < c^2$

Which of the following must be true?

**A** Statement I is true.

**B** Statement II is true.

**C** Both statements are true.

**D** At least one of the two statements is true.

**E** Exactly one of the two statements is true.

**F** Neither statement is true.

**END OF PAPER 1**

**Test of Mathematics for University Admission**

# TMUA October 2026

## Paper 2

Mathematical Reasoning

| | |
|---|---|
| **Time allowed** | 75 minutes |
| **Questions** | 20 |
| **Marks** | 20 |

**Instructions**

- Choose one answer for each question.
- Each correct answer is worth one mark. There is no penalty for an incorrect answer.
- Calculators and formula booklets must not be used.
- Use the space in this booklet for rough work.
- Record your responses in the Paper 2 section of the answer sheet.

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Unofficial reconstructed practice paper. See the compilation note at the beginning of Paper 1.

ducktmua.com

## Question 1

A fair six-sided die is tossed twice. Let $A_m$ be the event that the first toss lands on $m$, where $1 \leq m \leq 6$. Let $S_n$ be the event that the sum of the results is $n$, where $2 \leq n \leq 12$.

For how many pairs $(m, n)$ is

$$\mathrm{P}(A_m \text{ and } S_n) = \mathrm{P}(A_m) \times \mathrm{P}(S_n)\,?$$

**A** 0

**B** 1

**C** 6

**D** 11

**E** 30

**F** 36

**G** 66

## Question 2

For $x, y \in \mathbb{R}$ with $x > 1$ and $y > 1$, statements $P$ and $Q$ are defined as follows:

$$P: \quad \log_x y > 2,$$
$$Q: \quad \log_{xy}(y^3) > 2.$$

Which statement is true?

**A** $P$ is necessary but not sufficient for $Q$.

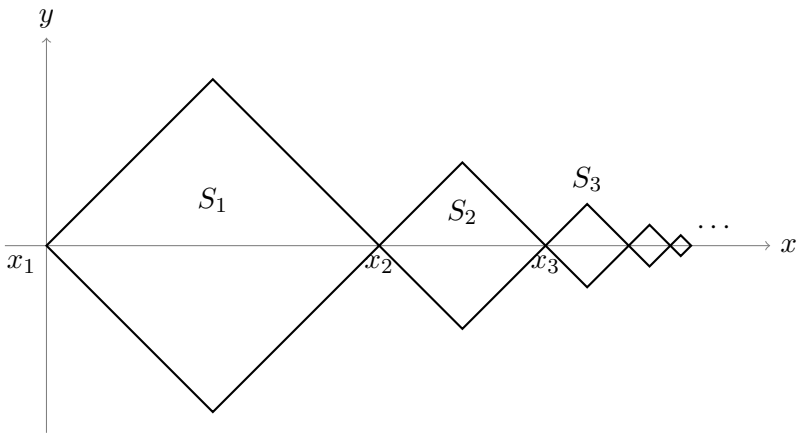
**B** $P$ is sufficient but not necessary for $Q$.

**C** $P$ is necessary and sufficient for $Q$.

**D** $P$ is neither necessary nor sufficient for $Q$.

## Question 3

The diagram shows infinitely many squares $S_n$, with $n$ increasing from left to right. The leftmost and rightmost vertices of $S_n$ have $x$-coordinates $x_n$ and $x_{n+1}$, respectively.




Given that

$$x_{n+1} = x_n + 2^{3-n}, \qquad x_1 = 0,$$

what is the sum of the areas of all the squares?

**A** 8

**B** $\frac{32}{3}$

**C** 16

**D** $\frac{64}{3}$

**E** $\frac{128}{3}$

**F** $\frac{8}{3}$

## Question 4

Which triple $(\ell, m, n)$, where $\ell$, $m$ and $n$ are all positive integers, provides a counterexample to the statement

"If $\ell \mid mn$, then $\ell \mid m$ or $\ell \mid n$" ?

**A** $\ell = 4, \quad m = 3, \quad n = 8$

**B** $\ell = 6, \quad m = 5, \quad n = 12$

**C** $\ell = 6, \quad m = 4, \quad n = 10$

**D** $\ell = 6, \quad m = 8, \quad n = 9$

## Question 5

Given that the statement

> There exists some real number $t$ (positive, negative or zero) such that
>
> $$\int_{-t}^{t} f(x)\,dx > 0$$

is false, which of statements I, II and III must be true? Assume that all the integrals below exist.

I There exists some real number $u$ such that $\int_{-u}^{u} f(x)\,dx \leq 0$.

II For all real numbers $v$, $\int_{-v}^{v} f(x)\,dx < 0$.

III $\int_{-1}^{1} f(x)\,dx = 0$.

**A** None of them

**B** I only

**C** II only

**D** III only

**E** I and II only

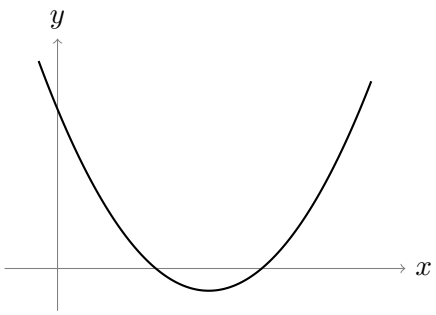
**F** I and III only

**G** II and III only

**H** I, II and III

## Question 6

The diagram shows the graph of $y = ax^2 + bx + c$, where $a$, $b$ and $c$ are non-zero real numbers.




Which of the following statements must be true?

I $b^2 - 4ac > 0$

II $abc > 0$

III $a + b + c > 0$

**A** None of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

## Question 7

The numbers $a$, $b$ and $c$ are positive integers. Consider the following statements:

$$P: \quad abc \text{ is divisible by } 4.$$
$$Q: \quad ab + bc + ac \text{ is even.}$$

Which of the following is true?

**A** $P$ is necessary but not sufficient for $Q$.

**B** $P$ is sufficient but not necessary for $Q$.

**C** $P$ is necessary and sufficient for $Q$.

**D** $P$ is neither necessary nor sufficient for $Q$.

## Question 8

Which of the following are square numbers?

I $63! \times 64!$

II $64!$

III $64! \times 65!$

**A** None of them

**B** I only

**C** II only

**D** III only

**E** I and II only

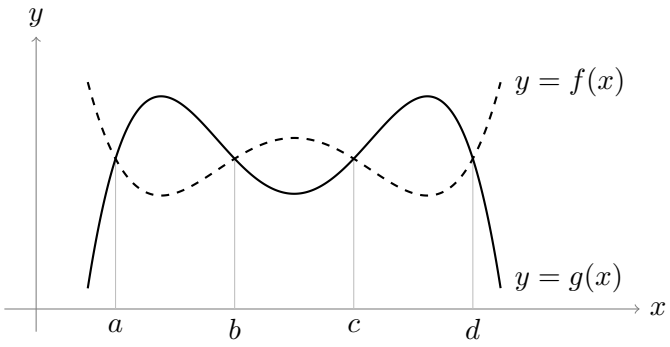
**F** I and III only

**G** II and III only

**H** I, II and III

## Question 9

The diagram shows the graphs of $y = f(x)$ (dashed for clarity) and $y = g(x)$. The two curves intersect at $x = a, b, c, d$ as shown and do not meet anywhere else.




Which of the following gives the area of the region bounded between the two curves?

**A** $\displaystyle\int_a^d \big(g(x) - f(x)\big)\,dx$

**B** $\displaystyle\int_a^d \big(f(x) - g(x)\big)\,dx$

**C** $\displaystyle\int_a^d \big(g(x) - f(x)\big)\,dx - 2\int_b^c \big(f(x) - g(x)\big)\,dx$

**D** $\displaystyle\int_a^d \big(f(x) - g(x)\big)\,dx + \int_b^c \big(g(x) - f(x)\big)\,dx$

**E** $\displaystyle\int_a^d \big(g(x) - f(x)\big)\,dx + \int_b^c \big(f(x) - g(x)\big)\,dx$

**F** $\displaystyle\int_a^d \big(g(x) - f(x)\big)\,dx + 2\int_b^c \big(f(x) - g(x)\big)\,dx$

**G** $\displaystyle\int_a^d \big(f(x) - g(x)\big)\,dx + 2\int_b^c \big(g(x) - f(x)\big)\,dx$

## Question 10

For a real number $p$, consider the following statements:

**S** $\cos x = -\sin(x - p)$ for all real $x$ if $p$ is an odd multiple of $\frac{\pi}{2}$.

**T** $\cos x = -\cos(x - p)$ for all real $x$ if and only if $p$ is an odd multiple of $\pi$.

Which of the following must be true?

I $S$

II The converse of $S$

III $T$

**A** None of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

## Question 11

A differentiable function satisfies

$$\int_0^x f(t)\,dt = x^3 - 2x^2 + x$$

for all real $x$.

Which of the following gives the complete range of values of $x$ for which $f'(x) > 0$?

**A** $x < \frac{1}{3}$ or $x > 1$

**B** $\frac{1}{3} < x < 1$

**C** $x < \frac{2}{3}$

**D** $x > \frac{3}{2}$

**E** $x > \frac{2}{3}$

**F** $x > \frac{1}{3}$

**G** $x > 1$

**H** No real value of $x$ satisfies the condition.

## Question 12

For some real number $k$, consider the lines with equations

$$y = 2^k x, \qquad y = x + k.$$

Which of the following best describes the behaviour of the two lines?

**A** The two lines have exactly one point of intersection for every real value of $k$.

**B** The two lines have no point of intersection for one value of $k$, and exactly one point of intersection for every other value of $k$.

**C** The two lines meet at infinitely many points for one value of $k$, and have exactly one point of intersection for every other value of $k$.

**D** The two lines have exactly one point of intersection when $k > 0$, and no point of intersection when $k \leq 0$.

## Question 13

Consider an arithmetic sequence with first term $a$, common difference $d$, number of terms $n$, and sum $S$, where $a$ and $d$ are integers and $n$ is a positive integer.

We are told whether $a$ is even or odd, whether $d$ is even or odd, and whether $n$ is even or odd. Which of the following statements is correct?

**A** We can always determine whether $S$ is even or odd.

**B** We cannot always determine whether $S$ is even or odd, but we can if the value of $n$ is also given.

**C** We cannot always determine whether $S$ is even or odd, but we can if the value of $a$ is also given.

**D** We cannot always determine whether $S$ is even or odd, but we can if the value of $d$ is also given.

## Question 14

All points on the curve $(x-p)^2+(y-q)^2=5$ satisfy $y+2x\leq 10$.

What is the maximum possible value of $2p+q$?

**A** $10-\sqrt{5}$

**B** $2\sqrt{5}$

**C** $\sqrt{5}$

**D** 5

**E** 10

## Question 15

Three distinct integers $a$, $b$ and $c$ are chosen from $1, 2, \ldots, 10$ and are such that exactly one of them is even and exactly one is a multiple of 3.

How many triples $(a, b, c)$ are possible? Order does not matter: for example, $(3, 1, 2)$ and $(3, 2, 1)$ count as the same triple.

**A** 15

**B** 27

**C** 50

**D** 66

**E** 68

**F** 162

## Question 16

Consider the infinite series

$$\frac{x}{1+|x|} + \frac{x}{(1+|x|)^2} + \frac{x}{(1+|x|)^3} + \cdots ,$$

where $x$ is real. Let its sum be $a$.

What is the complete set of values that $a$ can take?

**A** $\{-1, 0, 1\}$

**B** $\{-1, 1\}$

**C** $(-1, 1)$

**D** $[-1, 1]$

**E** $\{0, 1\}$

**F** $\{0\} \cup (1, \infty)$

## Question 17

A student attempts to solve the equation

$$(\log_2 x)^2 - 5\log_2 x + 6 = 0.$$

Their working is shown below:

I Rewrite the equation as $(\log_2 x)^2 - 5\log_2 x = -6$.

II Then $2\log_2 x - 5\log_2 x = -6$.

III So $\log_2 x = 2$.

IV Therefore $x = 4$.

Which of the following correctly describes the student's solution?

**A** The solution is completely correct.

**B** The solution is incorrect, and $x = 4$ is the unique solution to the equation.

**C** The solution is incorrect, and $x = 4$ is a solution to the equation but there is at least one other solution.

**D** The solution is incorrect, and $x = 4$ is not a solution to the equation.

## Question 18

A teacher writes the equation $x^2 + bx + c = 0$ on the board.

Student A copies $b$ correctly but $c$ incorrectly, obtaining an equation with roots $-4$ and $8$.

Student B copies $c$ correctly but $b$ incorrectly, obtaining an equation with roots $4$ and $-3$.

What are the roots of the teacher's equation?

**A** $4$ and $-3$

**B** $-4$ and $8$

**C** $6$ and $-2$

**D** $-6$ and $2$

**E** $12$ and $-4$

**F** The equation has no real roots.

## Question 19

A biased six-sided die has faces numbered 1 to 6.

The probability of rolling 1 is 0.2.

The probability of rolling an even number is 0.4.

The probability of rolling a prime number is 0.7.

What is the probability of rolling 2?

**A** 0.1

**B** 0.2

**C** 0.3

**D** 0.4

**E** 0.5

**F** There is insufficient information to determine the probability.

## Question 20

Consider the following conditions on real numbers $a$, $b$ and $c$:

I $2a > b$

II $a^2 + b^2 \leq c^2$

III $0 < a < b < c$

IV $a + c = 2b$

Which three conditions together guarantee that $a$, $b$ and $c$, in that order, form an arithmetic sequence and are the side lengths of a triangle?

**A** I, II and III

**B** I, II and IV

**C** I, III and IV

**D** II, III and IV

**E** None of these combinations

**END OF PAPER 2**

# Answer sheet

Write one answer letter in each box.

Name: _____ Date: _____

| Paper 1 | | Paper 2 | |
|---|---|---|---|
| Question | Answer | Question | Answer |
| 1 | | 1 | |
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